

The telephone operates via unique identification numbers that change from one location to another. Like a telegraph, the telephone connects geographical locations without the support of man-mobility. But unlike telegraph, it is interactive and supports real-time communication. It is therefore a step ahead of telegraph, but behind natural communication, as it does not support features such as multi-mode communication, fully wireless link, man-mobility and unique identification irrespective of location, among others.

Mobile communication—commonly known as cell telephony—was developed to infuse some sort of mobility in voice communication. To achieve this, geographic location-attached and portable handsets were replaced.

In mobile communication, portable handsets function using wireless (radio) links. Cellular or cell communication uses portable cellphones. The phones are registered in a geographic area called a cell. Like post offices, each of which have a geographical area of operation, cells are connected to cover wider geographical areas for communication.

This achieves the target of mobility. But this is primarily machine mobility, and not man-mobility of natural voice communication. Unless you carry the cellphone with you, mobile communication is not possible.

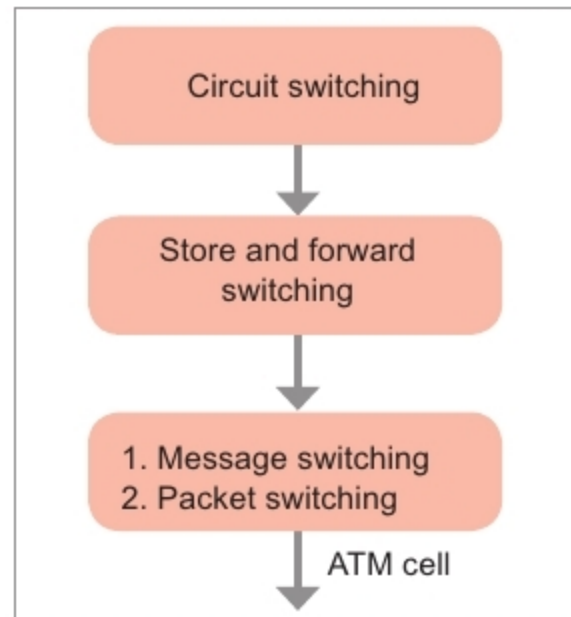


Figure 1: Virtual circuit services and datagram services

Several global standards are used for cellular communication. A few of these are GSM (Global Standard/Systems for Mobile), IS 95, 200 (international standard) and UMTS (Universal Mobile Telecommunication Systems).

With advancements in information technology such as integrated circuit (IC) technology, cell communication is now equipped with transportation of text messages, data, video along with interactive voice communication over a distance. Long-distance, interactive and real-time voice, video, text and data (hence, multimedia) communication over wireless links is now possible with mobile communication.

However, challenges of unique identification of communicating entities and man-mobility for natural voice

communication remain unresolved. With international roaming, of course, the issue of unique identification has been addressed to some extent.

Moving beyond cellphones, advancement of the Internet led to the support of multimedia real-time communication in many forms. The Internet is basically a data-carrying network. With the advent of VoIP (Voice over Internet Protocol), mobile IP and IPv6 (Internet Protocol version 6) technologies, the Internet can now carry multimedia over long distances, in no time.

The challenge of unique identification has been resolved further by the Internet. As an example, an e-mail ID can be used by only one unique person for communication. But, man-mobility still remains unresolved.

Personal communication network (PCN) was conceived, thereafter, to resolve the issue of man-mobility and unique identification for long-distance multimedia, interactive and long-haul communication. The idea first came about in the UK as a tool for a very-high-speed digital wireless cellular communication network with an intelligent system. Phones on the Move is a variation of personal communication, which is undergoing research in the UK. Development of PCN technology will pave the way for technology-enabled natural communication over long distance.

Data communication and networking

Stephen Hawking said, “We are all now connected by the Internet, like neurons in a giant brain.” This begs the question, why were we not connected by any of the predecessors of the Internet, such as telephony or telegraphy? The answer is, the Internet is a network that has converted BAD IT (where B stands for bursty, A for asymmetric, D for delicate and IT for insensitive to time) into an excellent network for all purposes.

In communication, all purposes cater to three basic services: voice,

Table 1: Some networks with standards of packets/frames

Network	Specification	Packet/frame size
Ethernet/LAN for office automation/IEEE 802.3	IEEE (Institute of Electrical & Electronics Engineers)	- Minimum: 64 bytes - Maximum: 1522 bytes of frame - Data/payload/packet size: 46 to 1500 bytes
Manufacturing Automation Protocol/IEEE 802.4	IEEE	- No minimum frame size - Maximum frame size: 8191 bytes
X.25	ITU-T (International Telecommunication Union)	128 to 256 bytes of packet
Synchronous data link control/High level data link control (SDLC/HDLC)	International Business Machines (IBM)	- Data/Payload/Packet: 46 to 1500 bytes - Overhead: 6 to 9 bytes for SDLC, and up to 20 bytes for HDLC